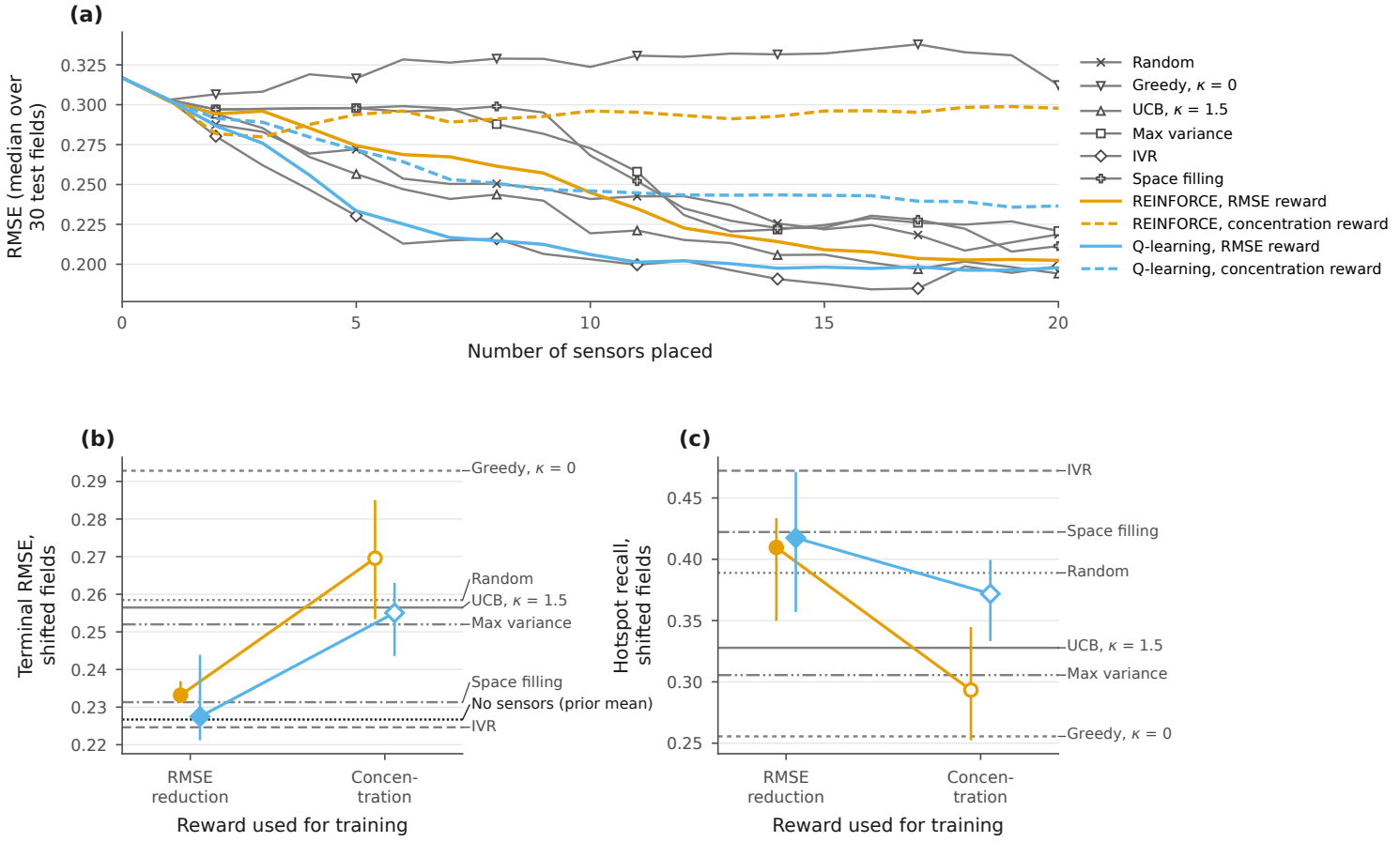
(a)
RMSE (median over 30 test fields)
0.325
0.300
0.275
0.250
0.225
0.200
0
5
10
15
20
Number of sensors placed
Random
Greedy, κ = 0
UCB, κ = 1.5
Max variance
IVR
Space filling
REINFORCE, RMSE reward
REINFORCE, concentration reward
Q-learning, RMSE reward
Q-learning, concentration reward
(b)
Terminal RMSE, shifted fields
0.29
0.28
0.27
0.26
0.25
0.24
0.23
0.22
Greedy, κ = 0
Random
UCB, κ = 1.5
Max variance
Space filling
No sensors (prior mean)
IVR
RMSE reduction
Concen-tration
Reward used for training
(c)
Hotspot recall, shifted fields
0.45
0.40
0.35
0.30
0.25
IVR
Space filling
Random
UCB, κ = 1.5
Max variance
Greedy, κ = 0
RMSE reduction
Concen-tration
Reward used for training